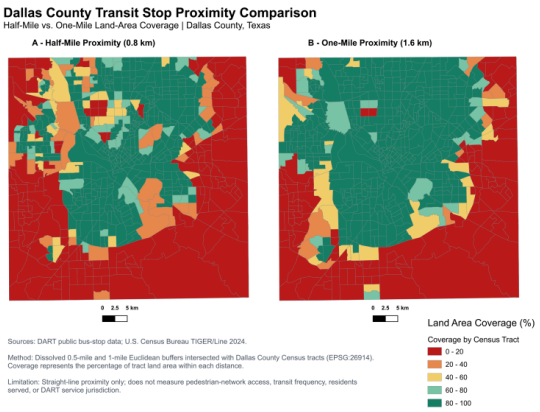
ArcGIS StoryMaps



Mapping Transit Accessibility in Dallas County

DART Bus-Stop Proximity, Household Vehicle Availability & Municipal Jurisdiction

Shweta Rane

October 27, 2026

Project introduction

How does proximity to public transit vary across Dallas County, and where might households without vehicles face additional transportation challenges?

This project combines DART bus-stop locations, 2024 U.S. Census tract boundaries, and 2020–2024 American Community Survey estimates to examine geographic proximity to transit stops and household vehicle availability.

Using QGIS, I analyzed 6,976 bus stops and 645 Census tracts, compared half-mile and one-mile proximity thresholds, and

identified 20 tracts meeting exploratory screening criteria for potential access gaps.

A subsequent municipal-boundary analysis distinguished flagged tracts inside and outside selected DART member-city areas, supporting a more geographically informed interpretation of the findings.

Research Objective

This study examines how geographic proximity to DART bus stops varies across Dallas County and whether areas with limited bus-stop proximity also have elevated shares of households without vehicles.

The analysis addresses three questions:

1. How does geographic coverage change between half-mile and one-mile proximity thresholds?
2. Which Census tracts combine low bus-stop proximity with higher estimated shares of zero-vehicle households?
3. How does accounting for DART member-city geography change the interpretation of potential access gaps?

The study uses straight-line distance as a preliminary geographic screening measure rather than a direct indicator of walking accessibility or transit service quality.

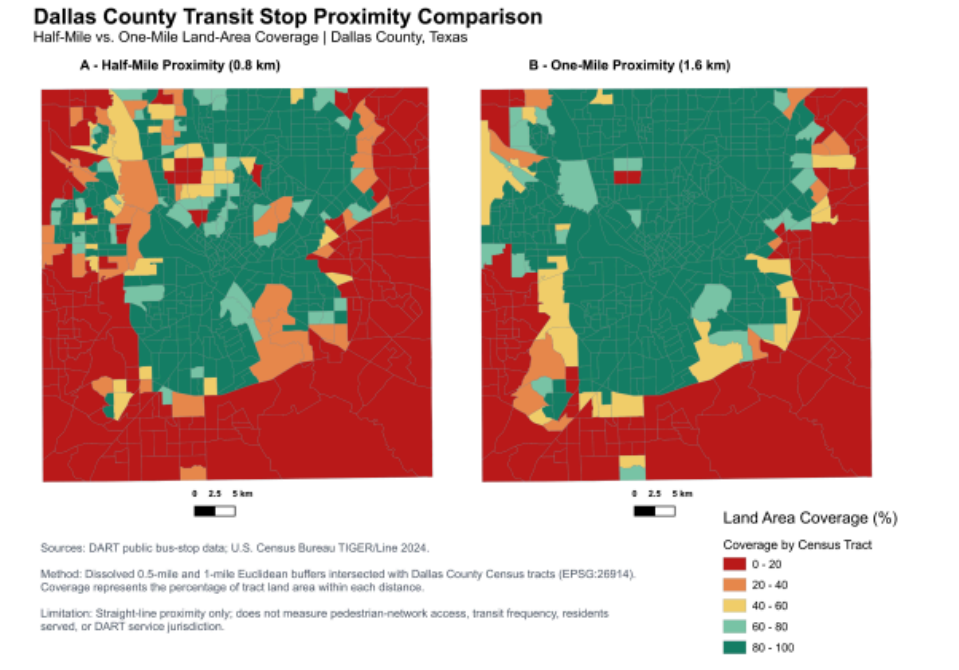


Figure 1.Comparison of half-mile and one-mile DART bus-stop proximity coverage. Both maps use consistent geographic scales and coverage classifications.

Data Sources & Methodology

Data Sources

This study integrates publicly available geospatial and demographic datasets:

- **DART Public Transit Data:** 6,976 bus-stop locations used to measure geographic proximity to transit infrastructure.
- **U.S. Census Bureau TIGER/Line 2024:** Census tract boundaries for 645 tracts in Dallas County, Texas.
- **American Community Survey (ACS) 2024 5-Year Estimates:** Table B08201, providing tract-level household vehicle-availability estimates and margins of error.
- **Texas Municipal Boundaries:** Geographic boundaries of selected DART member cities, used for a preliminary jurisdiction-overlap assessment.

Tools: QGIS, ArcGIS Online, ArcGIS StoryMaps, GeoPackage, and Census geographic identifiers (GEOIDs).

GIS Methodology

1. Data preparation: Standardized spatial datasets to NAD83 / UTM Zone 14N (EPSG:26914), validated geometry alignment, and joined Census demographic estimates to tract polygons using 11-digit GEOIDs.

2. Transit proximity analysis: Generated dissolved 0.5-mile (804.672 m) and 1-mile (1,609.344 m) straight-line buffers around DART bus stops, intersected them with Census tract polygons, and calculated the percentage of each tract's area within each distance.

3. Coverage comparison: Compared tract-level proximity coverage and calculated countywide area-weighted coverage using Census tract polygon areas.

4. Demographic screening: Calculated the estimated percentage of households without vehicles and flagged tracts with less than 40% half-mile proximity coverage and at least 10% estimated zero-vehicle households.

5. Municipal jurisdiction review: Intersected flagged tracts with selected DART member-city polygons and calculated geographic overlap to distinguish tracts fully outside, partially inside, and almost entirely inside member-city areas.

6. Quality assurance: Validated tract counts, geographic identifiers, joins, coverage percentages, spatial intersections, and classification totals.

Key Findings

Transit-stop proximity and household vehicle availability across 645 Dallas County Census tracts.

Metric	Result
Half-mile geographic coverage	**38.99%**
One-mile geographic coverage	**51.82%**
Increase in area coverage	**+12.83 percentage points**
Potential access-gap tracts	**20 of 645**
Estimated zero-vehicle households in flagged tracts	**4,033**

Expanding the proximity threshold from half a mile to one mile increased Dallas County's mapped tract-polygon area coverage by **12.83 percentage points**, revealing substantial differences in geographic coverage across distance thresholds.

The analysis identified **20 Census tracts** combining less than 40% half-mile bus-stop proximity coverage with an estimated zero-vehicle household share of at least 10%.

These tracts collectively contained an estimated **4,033 households without vehicles**. A subsequent municipal-boundary review found that **11 tracts had no overlap**, **4 had partial overlap**, and **5 had near-complete overlap** with the selected DART member-city geography.

These findings represent an exploratory geographic screening, not confirmed transit-service deficiencies.

Note: Coverage percentages represent area-weighted shares of mapped Census tract polygons, not percentages of residents served. Household counts

are ACS estimates subject to sampling uncertainty. Municipal overlap is a jurisdiction proxy, not a measure of actual transit availability.

1. Mapping Half-Mile Transit-Stop Proximity

To establish a baseline for geographic transit proximity, I generated half-mile (804.672 m) straight-line buffers around 6,976 mapped DART bus stops and calculated the percentage of each Dallas County Census tract polygon covered by these buffers.

The analysis reveals substantial spatial variation. Central areas generally show higher geographic coverage, while many peripheral tracts have lower proximity to the mapped bus stops.

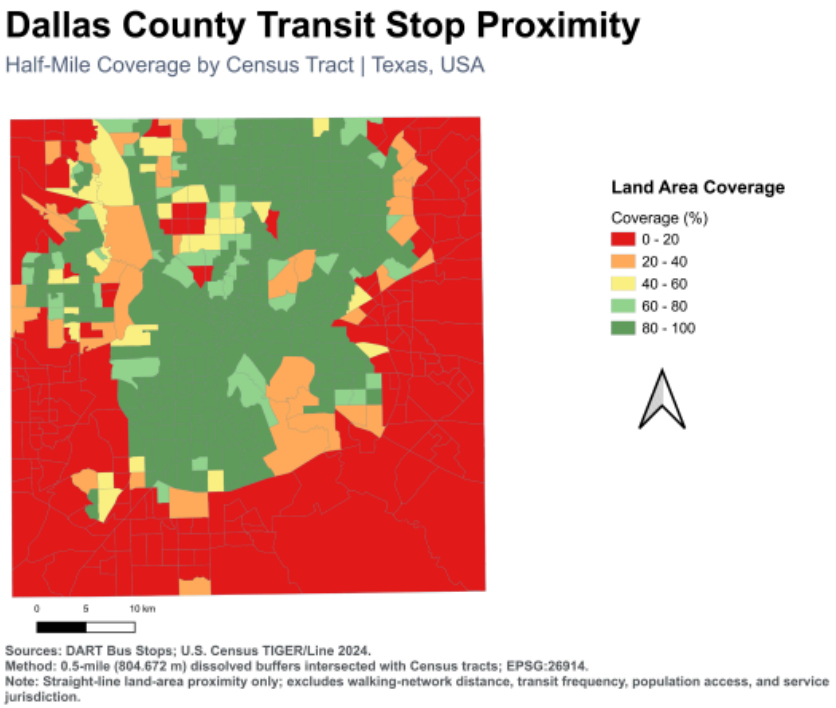


Figure 2. Half-mile straight-line proximity to DART bus stops across Dallas County Census tracts. Colors represent the percentage of each tract polygon within 804.672 meters of a mapped bus stop.

Finding: Approximately **38.99% of the total mapped Dallas County Census tract polygon area** falls within half a mile of a DART bus stop.

Although the unweighted mean tract coverage is 65.07%, the substantially lower area-weighted result indicates that larger tracts tend to have lower geographic proximity coverage.

This measure describes land-area proximity, not the share of residents within walking distance of transit.

2. Comparing Half-Mile and One-Mile Proximity

Expanding the straight-line proximity threshold from half a mile to one mile increases countywide area-weighted coverage from **38.99% to 51.82%**, a gain of **12.83 percentage points**.

However, the gains are unevenly distributed. Among 645 Census tracts, **335 show no reported percentage-point increase**, while **68 gain at least 40 percentage points**.

This comparison demonstrates how the selected distance threshold influences measured geographic accessibility.

As illustrated in **Figure 1 (presented in the Research Objective section above)**, expanding the proximity threshold from 0.5 mile to 1 mile increases geographic coverage across Dallas County. The side-by-side maps use consistent color classifications and geographic scales, allowing direct comparison of the spatial distribution of transit-stop proximity.

While overall mapped tract-polygon area coverage increases by **12.83 percentage points**, the improvement varies considerably across Census tracts. This highlights the importance of evaluating both countywide patterns and

localized differences when assessing geographic transit accessibility.

3. Household Vehicle Availability & Potential Access Gaps

To explore potential transportation vulnerability, I integrated 2024 American Community Survey (ACS) 5-Year household vehicle-availability estimates with the transit proximity results for 645 Dallas County Census tracts.

The analysis found that an average of **6.67% of households per tract** were estimated to have no vehicle available, with a median of **4.73%** across the 642 tracts with valid estimates.

Using exploratory thresholds of **less than 40% half-mile bus-stop proximity coverage** and **at least 10% estimated zero-vehicle households**, I identified **20 Census tracts** for further investigation.

Together, these tracts contained an estimated **4,033 households without vehicles**. This figure refers to households across the complete flagged tracts, not households individually verified as living beyond half a mile of a bus stop.

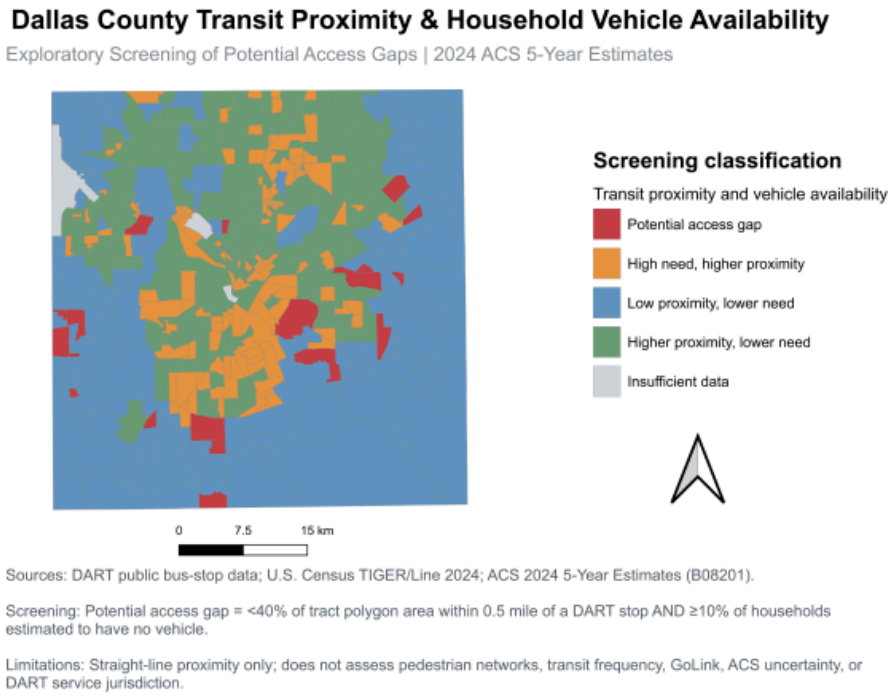


Figure 3. Household Vehicle Availability Screening. Census tracts classified by half-mile bus-stop proximity and estimated zero-vehicle household share. Red highlights the 20 tracts meeting the exploratory screening thresholds.

The screening revealed that low transit-stop proximity does not necessarily coincide with high estimated vehicle-related transportation need.

Of the 645 Census tracts analyzed, **20 (3.1%)** met both screening criteria. Another **127 tracts** had relatively high zero-vehicle household shares but greater geographic proximity to mapped bus stops, while **184 tracts** had low proximity but lower zero-vehicle household shares.

This distinction helps identify locations that may warrant closer investigation while avoiding the assumption that every area with limited bus-stop proximity experiences the same transportation challenges.

4. DART Member-City Jurisdiction Review

To account for differences in transit jurisdiction, I compared the 20 flagged Census tracts with municipal boundaries representing selected DART member cities.

The spatial overlay found that **11 flagged tracts had no overlap**, **4 had partial overlap**, and **5 had near-complete overlap** with the selected member-city polygons.

This additional check is important because limited proximity to DART bus stops outside member-city geography should not automatically be attributed to DART service deficiencies.

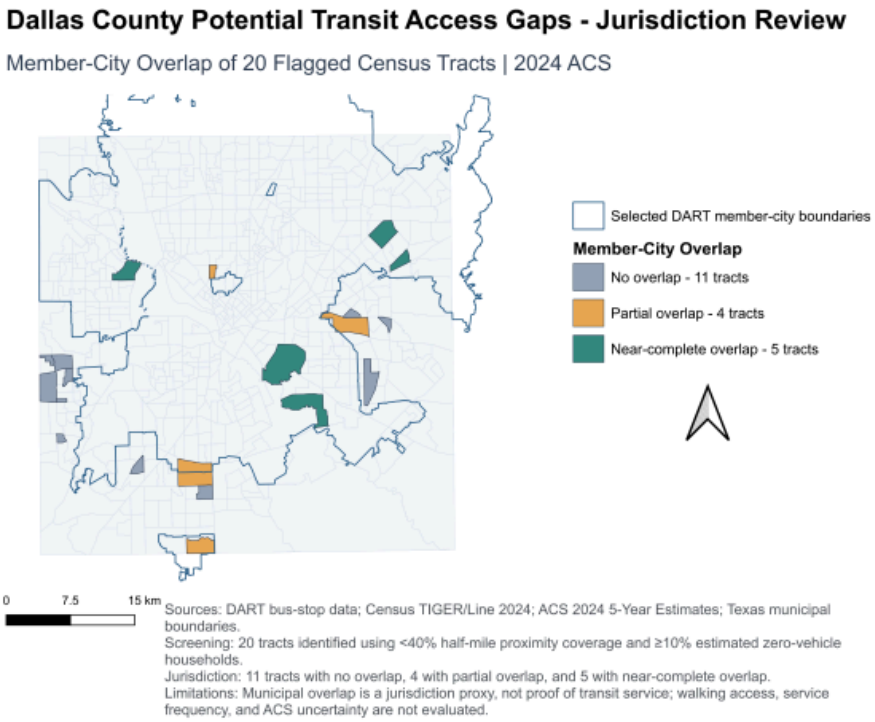


Figure 4. Jurisdiction Review. Geographic overlap between the 20 flagged Census tracts and selected DART member-city municipal boundaries. The overlay provides jurisdictional context but does not directly measure transit service availability.

Case Study: Census Tract 181.10

Census Tract 181.10 illustrates why jurisdictional context matters. The tract lies entirely within the selected DART member-city geography, but only **29.89% of its polygon area** is within half a mile of a mapped DART bus stop.

The 2024 ACS estimates **241 households without vehicles (12.15%)**, although the count has a substantial margin of error of **±236**.

This makes Tract 181.10 a candidate for closer examination of pedestrian access, transit routes, service frequency, and the location of residential areas—not evidence of a confirmed service deficiency.

5. Conclusions & Planning Implications

The analysis demonstrates that geographic proximity to public transit varies substantially across Dallas County, and that proximity alone does not fully capture transportation vulnerability.

Increasing the straight-line distance threshold from half a mile to one mile raised countywide mapped Census tract polygon-area coverage from **38.99% to 51.82%**, an increase of **12.83 percentage points**.

Combining transit proximity with ACS household vehicle-availability estimates identified **20 Census tracts** meeting the project's exploratory screening criteria, collectively containing an estimated **4,033 households without vehicles**.

However, municipal-boundary analysis revealed that **11 of the 20 flagged tracts had no overlap with the selected DART member-city geography**, while nine overlapped partially or nearly completely.

These findings illustrate why transportation planning assessments should consider geographic proximity, household characteristics, jurisdictional boundaries, and service conditions together. The flagged tracts provide candidates for further investigation rather than definitive evidence of transit inequity.

Potential Planning Implications

The results could support preliminary transportation planning by helping analysts:

- Identify Census tracts for detailed pedestrian-access and transit-service reviews.
- Prioritize further investigation where low bus-stop proximity overlaps elevated estimated zero-vehicle household shares.
- Distinguish potential DART-focused investigations from broader regional transportation concerns.
- Inform future analyses incorporating residential population distribution, walking networks, route frequency, and on-demand transit services.

6. Limitations & Future Work

Study Limitations

- **Straight-line distance:** Half-mile and one-mile buffers represent Euclidean proximity, not actual walking distance, pedestrian safety, or network accessibility.
- **Geographic aggregation:** Coverage is calculated as the percentage of each Census tract polygon's area near mapped bus stops. It does not measure the proportion of residents or households with transit access.
- **ACS uncertainty:** Household vehicle-availability figures are survey estimates with margins of error. Several flagged tracts have substantial uncertainty, limiting confident comparisons or rankings.
- **Transit service:** Bus-stop proximity does not capture service frequency, operating hours, route connectivity, rail access, GoLink, or transit reliability.
- **Municipal boundaries:** Selected DART member-city polygons provide preliminary jurisdictional context but are not authoritative measures of service availability.
- **Temporal alignment:** The analysis combines Census geography and ACS 2024 estimates with DART stop data and a later municipal-membership review. These sources do not necessarily represent a single historical snapshot.

Future Work

Future extensions could incorporate pedestrian-network service areas, population-weighted accessibility, ACS uncertainty-aware screening, current transit schedules and GTFS data, and verified DART jurisdiction boundaries. Incorporating household and employment locations would also support a more meaningful assessment of access to transportation opportunities.

7. Technical Skills Demonstrated

GIS & Spatial Analysis: QGIS, ArcGIS Online, ArcGIS StoryMaps, coordinate reference systems (EPSG:26914), Euclidean buffering, spatial intersections, dissolves, polygon area calculations, and graduated and categorized symbology.

Geospatial Data Engineering: GeoPackage, CSV ingestion, geographic identifier standardization, attribute joins, field calculations, spatial data validation, and quality assurance.

Demographic & Statistical Analysis: U.S. Census TIGER/Line, ACS 5-Year Estimates, area-weighted coverage calculations, descriptive statistics, household vehicle-availability screening, and interpretation of survey margins of error.

Geospatial Communication: Thematic cartography, comparative mapping, professional QGIS Print Layouts, geographic storytelling, and evidence-based presentation of findings and limitations.